

Arnav Pundir

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer proficient in building high-performance, real-time, and offline-first web systems. Experienced in React 19, Next.js 15, Node.js, Python (Flask), TypeScript, RxDB, and WebRTC. Strong expertise in engineering low-latency architectures, 3D WebGL performance, vector embeddings, state synchronization, and delivering production-grade client applications.

SKILLS

Languages : TypeScript, JavaScript, Python, C++, SQL, HTML5/CSS3
Frontend & WebGL : React 19, Next.js 15, Three.js (WebGL), GSAP, Tailwind CSS, RxDB (Offline PWA), Framer Motion
Backend & AI Systems : Node.js, Express.js, Flask, FastAPI, WebRTC, Redis, WebSockets, InsightFace (ArcFace), Gemini AI
Databases & Infrastructure : Supabase (PostgreSQL / pgvector), IndexedDB, MongoDB, Clerk Edge Auth, Technical SEO & Web Vitals, Docker, Git, Vercel

EDUCATION

COER University Roorkee, India
Bachelor of Technology in Computer Science and Engineering 2024 – 2028

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Developer Jan 2024 – Present
sparktech404.github.io (SparkTech Web Solutions) Roorkee, India

- Engineered commercial web platforms (e.g., SevaSetu) using **React.js** and REST APIs, translating specs into production UIs.
- Architected interactive 3D WebGL websites utilizing **Three.js** and **GSAP**, achieving **95+ Lighthouse performance scores**.
- Implemented component lazy-loading, Technical SEO (JSON-LD Schema, OpenGraph), and asset pipelines for client apps.

PROJECTS

ZeroLag Offline-First Platform | zero-lag-utms.vercel.app | github.com/ArnavPundir22/ZeroLag

- Architected a local-first task PWA using **React 19**, **TypeScript**, and **RxDB (IndexedDB)**, achieving **zero perceived UI latency** via optimistic client-side state updates.
- Built an event-sourced background sync engine with **Supabase PostgreSQL** and **Realtime WebSockets** for live presence updates.
- Integrated **Gemini Vision AI** via Vercel proxy to parse whiteboard photos into tasks, secured with **Clerk JWT + PostgreSQL RLS**.

BioSecure AI Biometric Platform | github.com/ArnavPundir22/Face-Attendance-System-Web-Version

- Deployed **InsightFace ArcFace 512D** face embeddings paired with **Supabase pgvector** search in PostgreSQL at **90%+ accuracy**.
- Engineered a 3D Pose-Gated EWMA Embedding Drift Engine ($\alpha = 0.30$) to monitor template aging and mitigate biometric degradation.
- Selected for official **COER University Pilot Run**; built a production **Flask** app with glassmorphic UI, SMTP alerts, and CSV exports.

AuraMeet Real-Time Video Chat | baatcheet-88e9.onrender.com | github.com/ArnavPundir22/BaatCheet

- Architected a peer-to-peer video streaming app using **Node.js**, **WebRTC**, and **Redis** delivering **sub-100ms low-latency** communication.
- Designed an ephemeral data lifecycle using **Redis pub/sub** that automatically purges chat logs and room state upon disconnect.
- Programmed a dynamic video canvas grid to optimize screen rendering performance across desktop and mobile devices.

LEADERSHIP & OPEN SOURCE CONTRIBUTIONS

Competitive Coding Committee Lead Jan 2025 – Present
dcoderssquad.com/leaders (Official Tech Club of COER University) Roorkee, India

- Spearhead algorithmic problem-solving workshops and mentor 100+ students in core Data Structures & Algorithms (DSA).
- Organize inter-departmental coding battles and co-led executive pilot sensitization sessions for BioSecure AI in front of university leadership.

Core Contributor Apr 2026 – Present
Nexus Spring of Code (NSoC '26) & GSSoC Global Rank #70

- Achieved **Global Rank #70** by merging **13 Pull Requests** across 5 open-source repositories, improving security and performance.
- Migrated legacy authentication handlers to secure, rotating **HttpOnly cookie sessions** and tightened CORS security policies.